

FINANCIAL INTELLIGENCE PLATFORM

Daily Progress Report

(Day 1 – Day 45)

Submitted By

Bhimishetti Lohith

Project Title

Financial Intelligence Platform

An End-to-End Financial Analytics and Business Intelligence System

Duration

45 Working Days

Technology Stack

- Python
- Pandas
- SQLite
- FastAPI
- Streamlit
- Plotly
- ReportLab
- Pytest
- Ruff
- Black

Table of Contents

1. Project Overview
2. Objectives
3. Technology Stack
4. Sprint 1 – Data Foundation (Day 1–7)

5. Sprint 2 – Financial Analytics (Day 8–14)
6. Sprint 3 – Business Intelligence (Day 15–21)
7. Sprint 4 – Advanced Analytics (Day 22–28)
8. Sprint 5 – Reporting & REST API (Day 29–35)
9. Sprint 6 – Documentation & Finalization (Day 36–45)
10. Final Summary
11. Lessons Learned
12. Future Enhancements
13. Conclusion

Project Overview

The Financial Intelligence Platform is an end-to-end financial analytics application developed to analyse historical financial statements of listed companies. The platform automates data ingestion, validates financial datasets, computes key financial ratios, performs company and sector analysis, provides investment screening capabilities, and generates professional reports.

The application combines data engineering, financial analytics, business intelligence, reporting, and visualization into a single integrated platform. Users can explore company performance through an interactive dashboard, compare peers, evaluate sectors, and generate analytical reports using historical financial data.

The project follows a structured software development lifecycle and is divided into six development sprints, each focusing on a specific phase of implementation.

Project Objectives

The primary objectives of the Financial Intelligence Platform are:

- Build an automated ETL pipeline for financial datasets.
 - Design a relational SQLite database for storing financial information.
 - Compute financial ratios and key performance indicators.
 - Develop an investment screening engine.
 - Provide company, peer, and sector analysis.
 - Generate professional PDF reports.
 - Develop REST APIs for data access.
 - Build an interactive Streamlit dashboard.
 - Produce comprehensive technical documentation.
-

Technology Stack

Category	Technologies
Programming Language	Python
Database	SQLite
Data Processing	Pandas, NumPy
Dashboard	Streamlit
API	FastAPI
Visualisation	Plotly
Reporting	ReportLab
Testing	Pytest
Code Quality	Ruff, Black
Version Control	Git, GitHub

Sprint 1 – Data Foundation

Sprint 1 focused on establishing the foundation of the project. The objective was to create the project structure, prepare the development environment, implement the ETL pipeline, validate input datasets, and load the processed data into the SQLite database.

Day 1

Objective

To initialise the project and establish the directory structure required for the Financial Intelligence Platform.

Activities Performed

- Created the project repository.
- Organised the directory structure.
- Created folders for ETL, analytics, dashboard, reports, API, tests, and documentation.
- Configured the Python virtual environment.
- Installed required Python packages.
- Prepared the initial project configuration.

Deliverables

- Project directory structure

- Virtual environment
- Dependency installation
- Initial Git repository

Outcome

The project environment was successfully configured, providing a structured foundation for subsequent development activities.

Day 2

Objective

To implement the Excel data loading mechanism.

Activities Performed

- Studied the financial datasets.
- Developed reusable Excel loading functions.
- Configured header normalization.
- Standardised file loading procedures.
- Tested data import functionality.

Deliverables

- Excel Loader
- Data loading utilities

Outcome

The ETL pipeline successfully imported financial datasets from Excel files into memory for further processing.

Day 3

Objective

To implement data normalization and validation.

Activities Performed

- Developed year normalization.
- Standardized company identifiers.
- Implemented validation rules.
- Tested duplicate detection.
- Performed initial data quality checks.

Deliverables

- Normalization module
- Validation module

Outcome

Input datasets were standardized, reducing inconsistencies before database loading.

Day 4**Objective**

To design the relational database schema.

Activities Performed

- Identified required entities.
- Designed relational tables.
- Defined primary keys.
- Defined foreign keys.
- Created SQL schema.

Deliverables

- Database schema
- SQL scripts

Outcome

The database structure was prepared to support efficient storage and querying of financial information.

Day 5**Objective**

To populate the SQLite database.

Activities Performed

- Connected ETL pipeline to SQLite.
- Loaded master data.
- Loaded financial statements.
- Implemented load auditing.
- Verified successful insertion.

Deliverables

- SQLite database
- Load audit report

Outcome

Financial data was successfully stored in the relational database for analytical processing.

Day 6**Objective**

To verify data quality and database integrity.

Activities Performed

- Validated row counts.
- Checked foreign key relationships.
- Reviewed random company records.
- Corrected inconsistencies.
- Improved validation reporting.

Deliverables

- Validation report
- Data quality report

Outcome

The database satisfied the required quality standards and was ready for financial analytics.

Day 7**Objective**

To complete Sprint 1 and prepare for financial analytics development.

Activities Performed

- Reviewed ETL pipeline.
- Optimized loader performance.
- Verified validation rules.
- Reviewed project structure.
- Prepared Sprint 2 implementation plan.

Deliverables

- Completed ETL pipeline

- Validated database
- Sprint review

Outcome

Sprint 1 concluded successfully with a fully functional ETL pipeline, validated financial datasets, and a structured relational database ready for analytical computations.

Sprint 1 Summary

Sprint 1 established the technical foundation of the Financial Intelligence Platform. By the end of the sprint, the project included a complete ETL workflow, standardized financial datasets, a relational SQLite database, and validation mechanisms that ensured data integrity. These components formed the basis for implementing the financial analytics engine in the next sprint.

Sprint 2 – Financial Analytics

Sprint 2 focused on developing the Financial Analytics Engine of the Financial Intelligence Platform. The objective was to transform raw financial statement data into meaningful business metrics by calculating financial ratios, growth indicators, profitability measures, liquidity analysis, leverage analysis, and financial scores.

The outputs of this sprint formed the analytical backbone of the application and were later used by the dashboard, screening engine, REST API, and reporting modules.

Day 8

Objective

To begin the implementation of the Financial Analytics Engine by identifying key financial ratios required for company analysis.

Activities Performed

- Reviewed financial statement structure.
- Identified important financial metrics.
- Categorized ratios into profitability, liquidity, leverage, efficiency, valuation, and growth.
- Planned the analytics module architecture.
- Created reusable calculation functions.

Deliverables

- Financial analytics module structure
- Ratio calculation plan
- KPI classification

Outcome

A structured framework for calculating financial ratios was prepared, ensuring consistency across all analytical modules.

Day 9

Objective

To implement profitability ratio calculations.

Activities Performed

- Calculated Return on Equity (ROE).
- Implemented Return on Assets (ROA).
- Computed Net Profit Margin.
- Calculated Operating Margin.
- Verified formulas against financial statements.

Deliverables

- Profitability ratio module
- Verified ratio calculations

Outcome

The platform successfully generated profitability metrics for companies using historical financial statement data.

Day 10

Objective

To implement liquidity and leverage ratio calculations.

Activities Performed

- Implemented Current Ratio.
- Calculated Quick Ratio.
- Computed Debt-to-Equity Ratio.
- Calculated Interest Coverage Ratio.
- Validated financial formulas.

Deliverables

- Liquidity ratio engine
- Leverage ratio engine

Outcome

The application successfully evaluated short-term liquidity and long-term financial stability of companies.

Day 11

Objective

To implement growth metrics and Compound Annual Growth Rate (CAGR) calculations.

Activities Performed

- Implemented Revenue CAGR.
- Calculated Profit CAGR.
- Computed EPS growth.
- Verified CAGR formulas using historical data.
- Tested growth calculations across multiple companies.

Deliverables

- CAGR module
- Growth analysis engine

Outcome

Historical growth performance of companies could now be measured accurately over multiple financial years.

Day 12

Objective

To generate a consolidated financial ratios dataset for all companies.

Activities Performed

- Combined all calculated financial ratios.
- Standardized output structure.
- Generated financial ratio records for every company.
- Stored computed metrics in the database.
- Validated output consistency.

Deliverables

- Financial ratios table
- Automated ratio generation pipeline

Outcome

A centralized repository of financial ratios was created for use by downstream analytical modules.

Day 13

Objective

To develop the financial scoring system.

Activities Performed

- Designed scoring criteria.
- Assigned weights to financial indicators.
- Implemented composite financial score calculation.
- Classified companies based on performance.
- Tested scoring accuracy.

Deliverables

- Financial scoring engine
- Company ranking logic

Outcome

Companies could now be evaluated using an overall financial score derived from multiple performance indicators.

Day 14

Objective

To complete Sprint 2 and validate all financial analytics.

Activities Performed

- Reviewed all ratio calculations.
- Performed validation against sample financial statements.
- Optimized analytical functions.
- Improved code modularity.
- Prepared analytical outputs for dashboard integration.

Deliverables

- Completed Financial Analytics Engine
- Validated ratio calculations

- Sprint review documentation

Outcome

Sprint 2 concluded successfully with a fully functional Financial Analytics Engine capable of generating profitability, liquidity, leverage, efficiency, valuation, and growth metrics for all companies in the dataset.

Sprint 2 Summary

Sprint 2 transformed raw financial statement data into meaningful analytical insights through the implementation of a comprehensive Financial Analytics Engine. Key financial ratios, growth indicators, and composite financial scores were developed and validated, enabling detailed evaluation of company performance.

The analytical outputs generated during this sprint formed the foundation for the dashboard, investment screener, peer comparison, sector analytics, REST API, and reporting features developed in subsequent sprints.

Sprint 3 – Business Intelligence & Dashboard Development

Sprint 3 focused on transforming the analytical outputs developed during Sprint 2 into meaningful business intelligence. The primary objective was to build interactive dashboard components, implement company profiling, develop peer and sector comparison modules, and create an investment screening engine that enables users to identify fundamentally strong companies.

This sprint significantly improved the usability of the Financial Intelligence Platform by providing an intuitive interface for financial analysis and decision-making.

Day 15

Objective

To initiate the development of the Streamlit dashboard and establish the user interface structure.

Activities Performed

- Designed the overall dashboard layout.
- Configured Streamlit application structure.
- Created navigation between dashboard pages.
- Developed reusable UI components.
- Established a consistent visual theme for the application.

Deliverables

- Dashboard framework
- Navigation menu
- User interface layout

Outcome

A structured dashboard interface was created, providing the foundation for integrating financial analytics into an interactive web application.

Day 16

Objective

To implement the Company Profile module for displaying detailed financial information of individual companies.

Activities Performed

- Designed the Company Profile page.
- Integrated company financial data from the database.
- Displayed key financial ratios and metrics.
- Added summary information and financial highlights.
- Verified data consistency between analytics and dashboard.

Deliverables

- Company Profile page
- Financial overview section
- Company analytics display

Outcome

Users could now explore detailed financial information and performance metrics for individual companies through the dashboard.

Day 17

Objective

To develop the Investment Screener for filtering companies based on financial criteria.

Activities Performed

- Defined screening parameters.
- Implemented filter logic for financial ratios.
- Created predefined screening presets.
- Enabled dynamic filtering based on user input.
- Validated screener results using financial data.

Deliverables

- Investment Screener module
- Screening presets
- Filter engine

Outcome

The platform enabled users to identify companies that satisfied predefined investment criteria, improving analytical efficiency.

Day 18

Objective

To implement the Peer Comparison module.

Activities Performed

- Identified peer groups.
- Compared financial ratios across companies.
- Generated comparative performance metrics.
- Designed graphical comparison views.
- Verified peer analytics calculations.

Deliverables

- Peer Comparison page
- Peer analytics engine
- Comparative financial visualizations

Outcome

Users could evaluate a company relative to its industry peers using multiple financial performance indicators.

Day 19

Objective

To implement Sector Analytics.

Activities Performed

- Grouped companies by sector.
- Calculated sector averages.
- Compared companies against sector benchmarks.
- Developed sector-wise visualizations.

- Verified analytical consistency.

Deliverables

- Sector Analytics page
- Sector benchmarking module
- Comparative charts

Outcome

The platform provided users with a broader understanding of company performance within the context of its industry sector.

Day 20

Objective

To integrate visualizations into the dashboard for improved financial interpretation.

Activities Performed

- Developed interactive charts using Plotly.
- Added trend visualizations.
- Integrated radar charts.
- Improved dashboard responsiveness.
- Optimized chart rendering performance.

Deliverables

- Interactive financial charts
- Trend analysis visualizations
- Dashboard enhancements

Outcome

Financial data became easier to interpret through interactive graphical representations, improving the overall user experience.

Day 21

Objective

To complete Sprint 3 by validating all business intelligence modules and preparing for advanced analytics.

Activities Performed

- Reviewed dashboard functionality.

- Validated company profile outputs.
- Tested investment screener.
- Verified peer and sector analytics.
- Improved navigation and user experience.

Deliverables

- Completed Business Intelligence dashboard
- Validated dashboard modules
- Sprint review documentation

Outcome

Sprint 3 concluded with a fully functional interactive dashboard capable of presenting company analytics, peer comparisons, sector benchmarks, and investment screening results through an intuitive user interface.

Sprint 3 Summary

Sprint 3 successfully transformed analytical outputs into business intelligence by integrating them into an interactive Streamlit dashboard. The Company Profile, Investment Screener, Peer Analytics, and Sector Analytics modules enabled users to explore financial information efficiently while providing meaningful insights through interactive visualizations.

The completion of Sprint 3 established a comprehensive user interface that served as the primary interaction layer for the Financial Intelligence Platform and laid the groundwork for implementing advanced analytics and reporting capabilities in subsequent sprints.

Sprint 4 – Advanced Financial Analytics

Sprint 4 focused on extending the analytical capabilities of the Financial Intelligence Platform by implementing advanced financial analysis modules. The primary objective was to provide deeper insights into company performance through Cash Flow Intelligence, capital allocation analysis, valuation metrics, financial trend evaluation, and enhanced visualizations.

These features enabled users to move beyond basic ratio analysis and better understand the financial strength, operational efficiency, and long-term sustainability of companies.

Day 22

Objective

To begin the implementation of the Cash Flow Intelligence module for analysing cash flow statements.

Activities Performed

- Studied the structure of cash flow statements.

- Identified key cash flow performance indicators.
- Designed reusable functions for cash flow analysis.
- Integrated cash flow data with existing financial records.
- Planned visualization of cash flow trends.

Deliverables

- Cash Flow Intelligence module
- Cash flow KPI framework
- Cash flow analysis functions

Outcome

The foundation for analysing operating, investing, and financing cash flows was successfully established.

Day 23**Objective**

To implement cash flow performance indicators and financial health assessment.

Activities Performed

- Calculated Operating Cash Flow.
- Computed Free Cash Flow.
- Implemented Cash Flow Margin.
- Analysed cash flow consistency across multiple years.
- Validated calculated values using financial statements.

Deliverables

- Cash flow KPI calculations
- Free Cash Flow analysis
- Financial health indicators

Outcome

The platform could now evaluate the cash-generating ability and operational sustainability of companies.

Day 24**Objective**

To develop the Capital Allocation analysis module.

Activities Performed

- Analysed capital expenditure trends.
- Evaluated dividend payments.
- Calculated retained earnings allocation.
- Reviewed debt repayment patterns.
- Generated capital allocation summaries.

Deliverables

- Capital Allocation module
- Capital expenditure analysis
- Dividend analysis

Outcome

Users gained insights into how companies allocate capital between business expansion, shareholder returns, and debt management.

Day 25

Objective

To implement valuation metrics for company assessment.

Activities Performed

- Calculated Price-to-Earnings (P/E) ratio.
- Implemented Price-to-Book (P/B) ratio.
- Computed Enterprise Value metrics.
- Reviewed valuation consistency.
- Integrated valuation indicators into the analytics engine.

Deliverables

- Valuation module
- Company valuation metrics
- Analytical calculations

Outcome

The Financial Intelligence Platform could now assess companies using commonly accepted valuation techniques.

Day 26

Objective

To improve financial trend analysis and historical performance visualization.

Activities Performed

- Developed trend analysis functions.
- Generated historical financial comparisons.
- Created trend charts for revenue, profit, and cash flow.
- Improved visualization performance.
- Verified analytical accuracy.

Deliverables

- Trend analysis module
- Historical financial charts
- Performance comparison tools

Outcome

Users could analyse long-term financial performance through interactive historical trend visualizations.

Day 27

Objective

To integrate advanced analytics with the dashboard.

Activities Performed

- Connected Cash Flow Intelligence with dashboard pages.
- Integrated valuation analysis.
- Displayed capital allocation summaries.
- Enhanced graphical presentation.
- Improved dashboard responsiveness.

Deliverables

- Advanced analytics dashboard
- Interactive visualizations
- Updated dashboard components

Outcome

The dashboard successfully incorporated advanced financial analytics, providing a richer user experience and more comprehensive financial insights.

Day 28

Objective

To complete Sprint 4 by validating advanced analytical modules.

Activities Performed

- Reviewed Cash Flow Intelligence outputs.
- Validated capital allocation calculations.
- Verified valuation metrics.
- Tested trend analysis visualizations.
- Prepared modules for reporting integration.

Deliverables

- Completed Advanced Analytics Engine
- Verified analytical modules
- Sprint review documentation

Outcome

Sprint 4 concluded with the successful implementation of advanced financial analytics, enabling deeper company evaluation through cash flow analysis, valuation metrics, capital allocation insights, and historical trend analysis.

Sprint 4 Summary

Sprint 4 significantly enhanced the Financial Intelligence Platform by introducing advanced analytical capabilities. Cash Flow Intelligence, capital allocation analysis, valuation metrics, and financial trend evaluation enabled users to perform comprehensive assessments of company performance beyond traditional ratio analysis.

These modules strengthened the analytical depth of the platform and prepared the system for automated report generation, REST API integration, and final project deployment in the subsequent sprint.

Sprint 5 – Reporting, REST API & System Integration

Sprint 5 focused on converting the analytical outputs into professional reports and exposing application functionality through REST APIs. During this sprint, automated PDF report generation, portfolio analysis, sector reports, FastAPI services, data export features, and overall system integration were implemented.

The objective was to make the Financial Intelligence Platform suitable for both end users and external applications by providing multiple ways to access financial insights.

Day 29

Objective

To begin the implementation of the reporting module for generating professional financial reports.

Activities Performed

- Studied report generation requirements.
- Designed report templates.
- Selected ReportLab for PDF generation.
- Defined report layout and formatting standards.
- Planned report automation workflow.

Deliverables

- Report generation framework
- PDF template design
- Reporting architecture

Outcome

A structured reporting framework was established for generating professional financial reports directly from analytical outputs.

Day 30**Objective**

To develop automated Company Tearsheets containing comprehensive financial summaries.

Activities Performed

- Designed Company Tearsheet layout.
- Included company overview information.
- Added financial ratio summaries.
- Integrated charts and performance indicators.
- Generated sample company reports.

Deliverables

- Company Tearsheet module
- Automated PDF generation
- Company summary reports

Outcome

The application successfully generated detailed PDF reports for individual companies, enabling users to review financial performance in a professional document format.

Day 31

Objective

To implement Portfolio Summary and Sector Reports.

Activities Performed

- Developed portfolio summary reports.
- Created sector-wise analytical reports.
- Included financial comparisons.
- Added graphical summaries.
- Standardized report formatting.

Deliverables

- Portfolio Summary Report
- Sector Report
- PDF reporting utilities

Outcome

Users could generate comprehensive reports summarizing portfolio performance and sector-level financial analysis.

Day 32

Objective

To implement REST APIs using FastAPI for exposing analytical data.

Activities Performed

- Configured FastAPI application.
- Developed API routing structure.
- Implemented health endpoint.
- Created company information endpoints.
- Designed ratio retrieval APIs.

Deliverables

- FastAPI application
- REST API endpoints
- API routing structure

Outcome

The platform exposed financial analytics through RESTful services, enabling integration with external applications.

Day 33

Objective

To extend REST API functionality and implement export features.

Activities Performed

- Developed screener API.
- Added peer analytics endpoints.
- Implemented sector analytics endpoints.
- Enabled CSV export functionality.
- Enabled Excel export functionality.

Deliverables

- Screener API
- Peer Analytics API
- Sector Analytics API
- Export utilities

Outcome

Users could retrieve analytical results programmatically while also exporting financial data for external analysis.

Day 34

Objective

To integrate all project modules into a unified application.

Activities Performed

- Connected dashboard with backend analytics.
- Integrated REST APIs with database.
- Linked report generation with dashboard.
- Verified data consistency across modules.
- Improved application performance.

Deliverables

- Fully integrated application
- Unified workflow
- End-to-end data pipeline

Outcome

All project modules functioned together as a single integrated Financial Intelligence Platform capable of supporting complete financial analysis workflows.

Day 35

Objective

To complete Sprint 5 through testing, validation, and system review.

Activities Performed

- Conducted end-to-end testing.
- Verified REST API functionality.
- Reviewed generated reports.
- Validated dashboard outputs.
- Identified and corrected integration issues.

Deliverables

- Tested reporting module
- Verified REST APIs
- Sprint review documentation

Outcome

Sprint 5 concluded successfully with a fully integrated application supporting interactive dashboards, automated reporting, REST APIs, and data export capabilities.

Sprint 5 Summary

Sprint 5 transformed the Financial Intelligence Platform into a complete analytical application by integrating reporting, REST API services, and export functionality. Automated Company Tearsheets, Portfolio Summary Reports, Sector Reports, and FastAPI endpoints significantly enhanced the usability and accessibility of the platform.

The successful completion of this sprint ensured that analytical outputs could be consumed through dashboards, downloadable reports, and programmatic interfaces, preparing the project for final documentation, code quality review, and acceptance activities in Sprint 6.

Sprint 6 – Documentation, Testing & Final Project Completion

Sprint 6 focused on completing the Financial Intelligence Platform through comprehensive documentation, code quality improvements, testing, validation, and final project packaging. The objective was to ensure that the application met project requirements, maintained high software quality standards, and was ready for deployment and evaluation.

This sprint involved preparing technical documentation, verifying application functionality, reviewing deliverables, and performing final acceptance checks before project submission.

Day 36

Objective

To review the overall project architecture and identify pending development tasks.

Activities Performed

- Reviewed all completed project modules.
- Verified folder organization.
- Checked integration between analytics, dashboard, API, and reporting modules.
- Identified pending documentation tasks.
- Prepared the final implementation roadmap.

Deliverables

- Project review checklist
- Architecture verification
- Sprint planning document

Outcome

The overall project structure was verified, and remaining work was clearly identified for successful project completion.

Day 37

Objective

To improve code quality and ensure maintainability.

Activities Performed

- Reviewed Python source files.
- Improved code readability.
- Refactored reusable functions.
- Standardized naming conventions.
- Removed redundant code segments.

Deliverables

- Refactored source code
- Improved project structure
- Code quality enhancements

Outcome

The project became more maintainable, modular, and easier to understand for future development.

Day 38**Objective**

To perform application testing and validation.

Activities Performed

- Executed unit tests.
- Validated database operations.
- Verified analytics calculations.
- Tested dashboard functionality.
- Confirmed REST API responses.

Deliverables

- Test execution report
- Validation results
- Bug fixes

Outcome

The application successfully passed functional testing, ensuring reliability across all major components.

Day 39**Objective**

To prepare comprehensive technical documentation.

Activities Performed

- Created the project README.
- Documented installation steps.
- Explained project architecture.
- Added workflow diagrams and folder descriptions.

- Prepared deployment instructions.

Deliverables

- README documentation
- Installation guide
- Technical documentation

Outcome

A complete technical guide was prepared to assist developers and evaluators in understanding and using the project.

Day 40**Objective**

To prepare the Analyst Guide and user documentation.

Activities Performed

- Documented dashboard usage.
- Explained report generation.
- Described REST API endpoints.
- Added troubleshooting guidelines.
- Included screenshots and workflow descriptions.

Deliverables

- Analyst Guide
- User manual
- Feature documentation

Outcome

End users received a comprehensive guide explaining how to operate and interpret the Financial Intelligence Platform.

Day 41**Objective**

To verify generated reports and analytical outputs.

Activities Performed

- Reviewed Company Tearsheets.
- Validated Portfolio Summary Reports.

- Checked Sector Reports.
- Verified chart rendering.
- Confirmed report formatting consistency.

Deliverables

- Verified PDF reports
- Reporting validation
- Quality review

Outcome

All generated reports were confirmed to be accurate, professionally formatted, and ready for distribution.

Day 42**Objective**

To conduct final project verification and acceptance testing.

Activities Performed

- Reviewed application workflow.
- Verified project deliverables.
- Checked documentation completeness.
- Confirmed module integration.
- Conducted acceptance review.

Deliverables

- Acceptance checklist
- Validation report
- Project verification

Outcome

The Financial Intelligence Platform successfully met the planned project objectives and passed the final verification process.

Day 43**Objective**

To prepare the project for submission.

Activities Performed

- Organized project directories.
- Verified required deliverables.
- Cleaned unnecessary files.
- Reviewed Git repository.
- Prepared submission packages.

Deliverables

- Submission-ready project
- Organized repository
- Final project package

Outcome

The project was successfully organized into a clean and professional submission structure.

Day 44**Objective**

To prepare the final documentation and completion reports.

Activities Performed

- Prepared the Day 44 Progress Report.
- Finalized the Analyst Guide.
- Reviewed Sprint documentation.
- Verified project README.
- Updated acceptance documentation.

Deliverables

- Day 44 Report
- Final Analyst Guide
- Updated README
- Acceptance Checklist

Outcome

All documentation was finalized, ensuring that the project met the required submission standards.

Day 45**Objective**

To complete the Financial Intelligence Platform and perform final project sign-off.

Activities Performed

- Conducted a final review of all project components.
- Verified dashboard, analytics engine, reporting module, and REST APIs.
- Confirmed technical documentation and project deliverables.
- Performed the final project sign-off.
- Prepared the project for evaluation and submission.

Deliverables

- Final Sign-Off Report
- Complete project documentation
- Final submission package
- Project acceptance documents

Outcome

The Financial Intelligence Platform was successfully completed and submitted as a fully integrated financial analytics application featuring automated ETL, financial analysis, business intelligence dashboards, professional PDF reporting, REST APIs, and comprehensive technical documentation.

Sprint 6 Summary

Sprint 6 completed the Financial Intelligence Platform by focusing on software quality, documentation, validation, and final project delivery. All modules were reviewed, tested, documented, and verified against project objectives.

The project reached production-ready quality with complete technical documentation, user guides, testing reports, acceptance documentation, and organized deliverables suitable for academic evaluation and professional presentation.

Overall Project Summary

The **Financial Intelligence Platform** was developed over **45 working days** following a structured six-sprint software development lifecycle.

Major accomplishments include:

- Automated ETL pipeline for financial datasets.
- Data validation and quality assurance.
- Relational SQLite database design.
- Financial ratio computation and analytics.
- Company, peer, and sector analysis.

- Investment screening engine.
- Interactive Streamlit dashboard.
- Professional PDF report generation.
- REST APIs using FastAPI.
- Comprehensive technical documentation.
- Automated testing and validation.
- Final project packaging and deployment readiness.

The project demonstrates practical application of **data engineering**, **financial analytics**, **business intelligence**, **software engineering**, and **full-stack Python development**.

Skills Acquired

During the development of this project, the following technical and professional skills were strengthened:

Technical Skills

- Python Programming
- Data Processing using Pandas
- SQLite Database Design
- ETL Pipeline Development
- Financial Ratio Analysis
- Business Intelligence
- Dashboard Development (Streamlit)
- REST API Development (FastAPI)
- PDF Report Generation
- Git & GitHub
- Unit Testing with Pytest
- Code Formatting and Quality Tools

Professional Skills

- Software Development Lifecycle (SDLC)
- Problem Solving
- Analytical Thinking
- Documentation
- Version Control

- Project Planning
 - Testing & Validation
 - Technical Communication
-

Challenges Faced

Throughout the project, several challenges were encountered, including:

- Handling inconsistent financial datasets.
- Designing reusable ETL components.
- Ensuring database integrity across multiple financial tables.
- Implementing accurate financial ratio calculations.
- Integrating analytics with interactive dashboards.
- Generating professional PDF reports.
- Maintaining modular and scalable code.
- Coordinating multiple project components into a unified application.

Each challenge was addressed through systematic testing, validation, and iterative improvements.

Future Enhancements

The Financial Intelligence Platform can be further enhanced by incorporating:

- Live stock market data integration.
 - Cloud database support (PostgreSQL/MySQL).
 - User authentication and role-based access.
 - Portfolio management features.
 - Machine learning models for stock prediction.
 - AI-assisted financial report summarization.
 - Mobile-responsive dashboard.
 - Cloud deployment using Docker and Kubernetes.
 - Real-time notifications and alerts.
 - Interactive forecasting and scenario analysis.
-

Conclusion

The Financial Intelligence Platform successfully achieved its objective of delivering an end-to-end financial analytics solution capable of transforming raw financial data into actionable business insights.

Through structured development across six sprints, the project integrates data engineering, financial analytics, visualization, reporting, REST APIs, and technical documentation into a single cohesive system. The completed platform demonstrates strong software engineering practices, modular architecture, and practical business intelligence capabilities, making it suitable for academic evaluation and as a portfolio project for industry opportunities.